

## II Shifting property

Find the Laplace transforms of the following

1.  $t^2 u(t-3)$

ans:  $\frac{e^{-3s}}{s^3} (9s^2 + 6s + 2)$

2.  $e^{-2t} u(t-2)$

ans:  $\frac{1}{e^2} \frac{e^{-s}}{(s+2)}$

3.  $\sin(t) u(t-\pi)$

ans:  $-\frac{e^{-\pi s}}{s^2 + 1}$